

ECON 300 – Labour Economics

Solutions: Final Exam Review

Winter 2026

Part I: Labour Supply and Labour Force Participation

1. Budget constraint with leisure and work

Given:

$$T = 112, \quad h = T - L, \quad w = 18, \quad V = 240$$

where V is non-labour income.

(a) Weekly budget constraint:

Weekly income C equals labour income plus non-labour income:

$$C = wh + V$$

Since $h = T - L$,

$$C = w(T - L) + V$$

Substitute the numbers:

$$C = 18(112 - L) + 240$$

$$C = 2016 - 18L + 240$$

$$\boxed{C = 2256 - 18L}$$

(b) Intercept on the income axis:

The income intercept occurs when leisure is zero:

$$L = 0$$

Then

$$C = 2256 - 18(0) = 2256$$

$$\boxed{2256}$$

(c) Intercept on the leisure axis:

The leisure intercept occurs when income equals non-labour income plus zero work, or more directly by setting hours worked to zero:

$$h = 0 \Rightarrow L = T = 112$$

So the leisure intercept is:

$$\boxed{112}$$

(d) Slope of the budget line:

From

$$C = 2256 - 18L$$

the slope is:

$$\boxed{-18}$$

Interpretation: one extra hour of leisure reduces income by \$18, which is the wage.

(e) If the wage rises to \$24:

New budget line:

$$C = 24(112 - L) + 240$$

$$C = 2688 - 24L + 240$$

$$\boxed{C = 2928 - 24L}$$

The new income intercept is:

$$C = 2928 \quad \text{when } L = 0$$

The leisure intercept remains:

$$L = 112$$

The slope becomes steeper:

$$\boxed{-24}$$

So the budget line rotates outward around the leisure intercept.

2. Hours worked before and after a wage increase

Initial leisure:

$$L_0 = 80$$

After the wage increase:

$$L_1 = 76$$

Total available time:

$$T = 112$$

(a) Hours worked before:

$$h_0 = T - L_0 = 112 - 80 = 32$$

Hours worked after:

$$h_1 = T - L_1 = 112 - 76 = 36$$

$$\boxed{h_0 = 32, \quad h_1 = 36}$$

(b) Labour supply increased because hours worked rose:

$$36 - 32 = 4$$

Labour supply increased by 4 hours

(c) Explanation:

A wage increase has two effects:

Substitution effect: Leisure becomes more expensive because each hour not worked now means giving up more income. This encourages the worker to consume less leisure and work more.

Income effect: A higher wage makes the worker richer for any given number of hours worked. If leisure is a normal good, the worker may demand more leisure and work less.

In this question, the worker reduced leisure from 80 to 76 hours, so the substitution effect dominated the income effect.

3. Reservation wage and labour force participation

A full workday is 8 hours. The worker values one hour of leisure at \$14. Fixed cost of working is \$60 per day.

(a) Minimum daily earnings needed to compensate for leisure forgone:

If the worker works 8 hours, leisure forgone is 8 hours. Value of forgone leisure:

$$8 \times 14 = 112$$

\$112

(b) Including fixed costs:

Total daily compensation required:

$$112 + 60 = 172$$

\$172

(c) Reservation wage per hour:

Divide total required daily earnings by 8 hours:

$$\frac{172}{8} = 21.5$$

\$21.50 per hour

(d) If market wage is \$21:

Daily earnings would be:

$$21 \times 8 = 168$$

Since

$$168 < 172$$

the worker will not participate.

No, the person will not participate

Part II: Income Maintenance, Welfare, and Work Incentives

4. Income maintenance program

Given:

$$G = 700, \quad t = 0.40, \quad w = 25$$

(a) Disposable income while benefits are positive:

The standard formula is:

$$C = G + (1 - t)wh$$

Substitute:

$$C = 700 + (1 - 0.40)(25)h$$

$$C = 700 + 0.60(25)h$$

$$C = 700 + 15h$$

(b) Effective net wage:

The effective net wage is:

$$(1 - t)w = 0.60(25) = 15$$

$$\text{\$15 per hour}$$

(c) Break-even level of earnings:

Benefits fall to zero when:

$$G = twh$$

or equivalently when gross earnings satisfy:

$$wh = \frac{G}{t}$$

So:

$$wh = \frac{700}{0.40} = 1750$$

$$\text{\$1750}$$

(d) Corresponding hours of work:

Since $w = 25$,

$$25h = 1750$$

$$h = 70$$

$$\boxed{70 \text{ hours}}$$

(e) Effect of higher benefit reduction rate:

A higher tax-back rate lowers the effective net wage:

$$(1 - t)w$$

so workers keep less of each extra dollar earned. This weakens the incentive to work additional hours and may reduce labour supply.

5. Demogrant vs. negative income tax style program

Wage:

$$w = 20$$

(a) Program A: demogrant of \$500

A demogrant is a lump-sum transfer with no tax-back, so:

$$C = 500 + 20h$$

$$\boxed{C = 500 + 20h}$$

(b) Program B: guaranteed income of \$500 with tax-back rate 50%

Disposable income while benefits are positive:

$$C = 500 + (1 - 0.5)(20)h$$

$$C = 500 + 10h$$

$$\boxed{C = 500 + 10h}$$

(c) Effective net wage under Program B:

$$(1 - 0.5)20 = 10$$

$$\boxed{\$10}$$

(d) Which program creates a substitution effect?

Program A does *not* change the wage rate, so it creates only an income effect.

Program B lowers the effective wage from \$20 to \$10, so it creates a substitution effect.

Program B creates a substitution effect

(e) Which program is more likely to reduce hours worked?

Program A raises income without changing the wage, so it may reduce hours through an income effect.

Program B raises income and also lowers the effective wage, so it reduces work incentives more strongly because it creates both an income effect and a substitution effect.

Program B is more likely to reduce hours worked

Part III: Labour Demand in Competitive Markets

6. Competitive firm labour demand

Production:

$$Q = 40N - N^2$$

Output price:

$$P = 3$$

Wage:

$$w = 48$$

(a) Marginal product of labour:

Differentiate Q with respect to N :

$$MP_N = \frac{dQ}{dN} = 40 - 2N$$

$$MP_N = 40 - 2N$$

(b) Value of marginal product:

$$VMP_N = P \cdot MP_N = 3(40 - 2N) = 120 - 6N$$

$$VMP_N = 120 - 6N$$

- (c) Profit-maximizing employment:
Competitive firms hire labour until:

$$VMP_N = w$$

So:

$$120 - 6N = 48$$

$$72 = 6N$$

$$N = 12$$

$$\boxed{N = 12}$$

- (d) Total output at $N = 12$:

$$Q = 40(12) - 12^2 = 480 - 144 = 336$$

$$\boxed{Q = 336}$$

7. Elasticity of labour demand

Given:

$$\eta = -1.4, \quad N_0 = 250, \quad \% \Delta W = 10\%$$

- (a) Percentage change in employment:

By definition:

$$\% \Delta N = \eta \cdot \% \Delta W$$

$$\% \Delta N = -1.4(10\%) = -14\%$$

$$\boxed{-14\%}$$

- (b) New employment:

A 14% decrease from 250:

$$N_1 = 250(1 - 0.14) = 250(0.86) = 215$$

$$\boxed{215}$$

- (c) Meaning of elastic labour demand:

Labour demand is elastic when employment responds strongly to wage changes. Here, a 10% wage increase causes a 14% fall in employment, so firms are quite responsive to wage changes.

8. Cost-minimizing input choice

Given:

$$MPP_N = 20, \quad MPP_K = 10, \quad w = 40, \quad r = 15$$

(a)

$$\frac{MPP_N}{MPP_K} = \frac{20}{10} = 2$$

2

(b)

$$\frac{w}{r} = \frac{40}{15} = 2.67$$

2.67

(c) Cost minimization requires:

$$\frac{MPP_N}{MPP_K} = \frac{w}{r}$$

Since

$$2 \neq 2.67$$

the condition is not satisfied.

No

(d) Since

$$\frac{MPP_N}{MPP_K} < \frac{w}{r}$$

labour is relatively expensive compared with its productivity contribution. The firm should use less labour and more capital.

Use relatively less labour and more capital

Part IV: Monopsony and Wage Determination

9. Monopsony equilibrium

Supply:

$$W = 12 + 0.5N$$

Demand:

$$MRP_N = 60 - 0.5N$$

(a) Marginal cost of labour:

Total labour cost:

$$TLC = W \cdot N = (12 + 0.5N)N = 12N + 0.5N^2$$

Differentiate:

$$MCL = \frac{dTLC}{dN} = 12 + N$$

$$\boxed{MCL = 12 + N}$$

(b) Monopsony employment:

Set

$$MCL = MRP_N$$

$$12 + N = 60 - 0.5N$$

$$1.5N = 48$$

$$N = 32$$

$$\boxed{N_M = 32}$$

(c) Monopsony wage:

Substitute into supply:

$$W = 12 + 0.5(32) = 12 + 16 = 28$$

$$\boxed{W_M = 28}$$

(d) Competitive outcome:

Under competition:

$$W = MRP_N$$

so

$$12 + 0.5N = 60 - 0.5N$$

$$N = 48$$

Wage:

$$W = 12 + 0.5(48) = 36$$

$$\boxed{N_C = 48, \quad W_C = 36}$$

(e) Comparison:

Monopsony leads to:

$$N_M = 32 < 48 = N_C$$

$$W_M = 28 < 36 = W_C$$

So employment and wages are both lower under monopsony.

10. Minimum wage under monopsony

Minimum wage:

$$W_{min} = 32$$

(a) Maximum employment available from supply curve:

From

$$32 = 12 + 0.5N$$

$$20 = 0.5N$$

$$N = 40$$

40

(b) Firm's desired employment:

With a binding minimum wage, marginal cost is 32 up to $N = 40$. Set:

$$32 = 60 - 0.5N$$

$$0.5N = 28$$

$$N = 56$$

But only 40 workers are willing to work at that wage, so the feasible employment is:

40

(c) Compare with original monopsony outcome:

Original monopsony employment:

$$32$$

New employment:

$$40$$

Employment rises

(d) Explanation:

Under monopsony, the firm hires too few workers because hiring more labour drives up wages. A minimum wage can flatten marginal labour cost over some range, allowing the firm to hire more workers and move closer to the competitive outcome.

Part V: Compensating Wage Differentials

11. Risky vs safe jobs

Safe job:

\$28

Risky job:

\$34

(a) Compensating differential:

$$34 - 28 = 6$$

\$6 per hour

(b) Worker requires at least \$5:

Since

$$6 > 5$$

the risky job compensates enough.

Choose risky job

(c) Worker requires at least \$7:

Since

$$6 < 7$$

the premium is not enough.

Choose safe job

(d) Sorting explanation:

Workers differ in how much compensation they require for unpleasant or risky job characteristics. More risk-tolerant workers accept risky jobs at lower premia, while more risk-averse workers choose safer jobs unless wage compensation is very high.

12. Value of a statistical life

Annual premium:

\$1,200

Increase in fatality risk:

$$\frac{1}{20,000}$$

(a) VSL:

$$VSL = \frac{1200}{1/20,000} = 1200(20,000) = 24,000,000$$

$$\boxed{\$24,000,000}$$

(b) Interpretation:

This does not mean any specific life is “worth” \$24 million. It means that, aggregated across many workers, the labour market reveals a willingness to pay consistent with \$24 million for reducing one expected statistical death.

Part VI: Human Capital, Education, and Training

13. Johnny’s diploma decision

Annual direct costs:

$$2500$$

Forgone earnings:

$$18,000$$

Earnings without diploma:

$$30,000$$

Earnings with diploma:

$$38,000$$

Discount rate:

$$r = 0.05$$

Working life after graduation:

$$T = 20$$

(a) Total annual cost:

$$2500 + 18,000 = 20,500$$

$$\boxed{\$20,500}$$

(b) Present value of 3 years of costs:

Costs are paid at the end of each year:

$$PV_C = \frac{20,500}{1.05} + \frac{20,500}{(1.05)^2} + \frac{20,500}{(1.05)^3}$$

Compute each term:

$$\frac{20,500}{1.05} = 19,523.81$$

$$\frac{20,500}{1.1025} = 18,594.10$$

$$\frac{20,500}{1.157625} = 17,708.67$$

Sum:

$$PV_C = 19,523.81 + 18,594.10 + 17,708.67 = 55,826.58$$

$$PV_C \approx \$55,826.58$$

(c) Annual earnings gain:

$$38,000 - 30,000 = 8,000$$

$$\$8,000$$

(d) Present value of benefits at time of graduation:

Use annuity formula:

$$PV_B^{(3)} = 8000 \left(\frac{1 - (1.05)^{-20}}{0.05} \right)$$

Since

$$(1.05)^{-20} \approx 0.37689$$

then

$$\frac{1 - 0.37689}{0.05} = 12.4622$$

so

$$PV_B^{(3)} = 8000(12.4622) = 99,697.60$$

$$PV_B^{(3)} \approx \$99,697.60$$

(e) Discount benefits back to the present:

Graduation is after 3 years, so:

$$PV_B^{(0)} = \frac{99,697.60}{(1.05)^3}$$

$$PV_B^{(0)} = \frac{99,697.60}{1.157625} \approx 86,122.02$$

$$PV_B^{(0)} \approx \$86,122.02$$

(f) NPV rule:

$$NPV = PV_B^{(0)} - PV_C$$
$$NPV = 86,122.02 - 55,826.58 = 30,295.44$$

Since $NPV > 0$, Johnny should invest.

Yes, Johnny should invest in the diploma

14. General vs specific training

- (a) General training is more likely financed by the worker because it increases earnings in many firms, so the worker captures most of the benefit.
- (b) Specific training is more likely financed by the firm because the productivity gain is mainly valuable inside the current firm.

(c) Why it matters for wages:

During general training, trainees may accept lower wages because they will gain future transferable benefits. Under specific training, firms often share more of the cost because the future gains are not fully portable.

Part VII: Discrimination and Oaxaca Decomposition

15. Oaxaca decomposition

Men:

$$\ln W_m = 1.10 + 0.08S + 0.05EXP$$

Women:

$$\ln W_f = 0.90 + 0.07S + 0.04EXP$$

Means:

$$\bar{S}_m = 16, \quad E\bar{X}P_m = 12$$

$$\bar{S}_f = 15, \quad E\bar{X}P_f = 10$$

(a) Predicted log earnings for men:

$$\begin{aligned}\ln W_m &= 1.10 + 0.08(16) + 0.05(12) \\ &= 1.10 + 1.28 + 0.60 = 2.98\end{aligned}$$

2.98

(b) Predicted log earnings for women:

$$\begin{aligned}\ln W_f &= 0.90 + 0.07(15) + 0.04(10) \\ &= 0.90 + 1.05 + 0.40 = 2.35\end{aligned}$$

$$\boxed{2.35}$$

(c) Total log wage gap:

$$2.98 - 2.35 = 0.63$$

$$\boxed{0.63}$$

(d) Explained portion using male structure:

Difference in schooling:

$$16 - 15 = 1$$

Difference in experience:

$$12 - 10 = 2$$

Multiply by male coefficients:

$$(1)(0.08) + (2)(0.05) = 0.08 + 0.10 = 0.18$$

$$\boxed{0.18}$$

(e) Unexplained portion:

$$0.63 - 0.18 = 0.45$$

$$\boxed{0.45}$$

(f) Interpretation:

Of the total log wage gap of 0.63, 0.18 is explained by measured differences in schooling and experience. The remaining 0.45 is unexplained and may reflect discrimination or omitted variables.

Part VIII: Immigration

16. Immigration categories

Total:

$$360,000$$

(a) Shares:

Economic:

$$\frac{180,000}{360,000} \times 100 = 50\%$$

Family:

$$\frac{108,000}{360,000} \times 100 = 30\%$$

Refugees:

$$\frac{72,000}{360,000} \times 100 = 20\%$$

$$\boxed{50\%, 30\%, 20\%}$$

(b) Economic immigrants rise by 15%:

$$180,000(1.15) = 207,000$$

New total:

$$207,000 + 108,000 + 72,000 = 387,000$$

$$\boxed{387,000}$$

(c) New economic share:

$$\frac{207,000}{387,000} \times 100 \approx 53.49\%$$

$$\boxed{53.49\%}$$

17. Migration decision

Earnings:

$$54,000 - 48,000 = 6,000$$

So the annual advantage is:

$$\boxed{\$6,000}$$

(a) Annual advantage:

$$\boxed{\$6,000}$$

(b) Present value:

$$PV = 6000 \left(\frac{1 - (1.04)^{-25}}{0.04} \right)$$

Since

$$(1.04)^{-25} \approx 0.3751$$

then

$$\frac{1 - 0.3751}{0.04} = 15.6225$$

so

$$PV = 6000(15.6225) = 93,735$$

$$\boxed{\$93,735}$$

(c) Subtract moving cost:

$$93,735 - 20,000 = 73,735$$

Since this is positive, the migrant should move.

$$\boxed{\text{Yes, the migrant should move}}$$

Part IX: Unemployment Measurement and Dynamics

18. Basic labour market indicators

Population:

$$8,000,000$$

Employed:

$$5,360,000$$

Unemployed:

$$340,000$$

(a) Labour force:

$$5,360,000 + 340,000 = 5,700,000$$

$$\boxed{5,700,000}$$

(b) Unemployment rate:

$$\frac{340,000}{5,700,000} \times 100 = 5.96\%$$

$$\boxed{5.96\%}$$

(c) Labour force participation rate:

$$\frac{5,700,000}{8,000,000} \times 100 = 71.25\%$$

$$\boxed{71.25\%}$$

(d) Employment rate:

$$\frac{5,360,000}{8,000,000} \times 100 = 67.00\%$$

67.00%

19. Changes in unemployment

Labour force:

2,500,000

Unemployment rate:

8.4%

(a) Unemployed:

$$0.084(2,500,000) = 210,000$$

210,000

(b) Employed:

$$2,500,000 - 210,000 = 2,290,000$$

2,290,000

(c) New unemployed workers:

35,000 unemployed find jobs:

$$210,000 - 35,000 = 175,000$$

20,000 employed workers lose jobs and search:

$$175,000 + 20,000 = 195,000$$

195,000

(d) New unemployment rate:

$$\frac{195,000}{2,500,000} \times 100 = 7.80\%$$

7.80%

20. Incidence and duration

(a) Region A:

$$UR = 3\% \times 2.5 = 7.5\%$$

Region B:

$$UR = 1.5\% \times 5 = 7.5\%$$

$$\boxed{UR_A = 7.5\%, \quad UR_B = 7.5\%}$$

(b) Higher incidence:

$$3\% > 1.5\%$$

Region A

(c) Longer duration:

$$5 > 2.5$$

Region B

(d) Policy explanation:

The same unemployment rate can come from many short unemployment spells or fewer long spells. Region A's problem is frequent entry into unemployment. Region B's problem is prolonged joblessness. These require different policies.